V4A1 – ALGEBRAIC GEOMETRY I
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Preliminaries

These notes roughly correspond to the course **V4A1 – Algebraic Geometry I** taught by Prof. Daniel Huybrechts at the Universität Bonn in the Winter 2024/25 semester. These notes are LaTeX-ed after the fact with significant alteration and are subject to misinterpretation and mistranscription. Use with caution. Any errors are undoubtedly my own and any virtues that could be ascribed to these notes ought be attributed to the instructor and not the typist. Knowledge of commutative algebra, topology, and category theory will be assumed.

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1. Lecture 1 – 7th October 2024

All rings are to be taken as commutative.

This will be a lecture course on algebraic geometry based on the abstract perspective of sheaves and schemes. The basic construction to be introduced is sheaves, which are defined as presheaves that satisfy additional properties. Sheaves keep track of both local and global information on topological spaces. We first consider the case of presheaves.

Sheaves can in fact be defined more abstractly using sites and Grothendieck topologies.

Definition 1.1 (Presheaf)**.** Let C be a category and X a topological space. A C-valued presheaf $\mathcal{F}$ on X consists of the data of

(i) An object $\mathcal{F}(U)$ of C for each $U \subseteq X$ open.
(ii) A morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ in C for each $V \subseteq U \subseteq X$ open.

such that $\mathrm{res}_{U,U} : \mathcal{F}(U) \to \mathcal{F}(U)$ is the identity on $\mathcal{F}(U)$ and the triangle

$$\begin{tikzcd} \mathcal{F}(U) \arrow[rr, "\mathrm{res}_{U,V}"] \arrow[dr, "\mathrm{res}_{U,W}"'] & & \mathcal{F}(V) \arrow[dl, "\mathrm{res}_{V,W}"] \\ & \mathcal{F}(W) & \end{tikzcd} \tag{1.1}$$

is commutative for all $W \subseteq V \subseteq U \subseteq X$ open.

Remark 1.2. Commutativity of the triangle is equivalent to $\mathrm{res}_{U,W} = \mathrm{res}_{V,W} \circ \mathrm{res}_{U,V}$ as morphisms in C.

We will typically consider examples where C is the category of Abelian groups AbGrp, of rings Ring, of modules over a fixed ring A Mod_A and their appropriate generalizations in the setting of schemes. While it is typically bad categorical practice to look "within" objects of a category, we refer to elements of $\mathcal{F}(U)$ for $U \subseteq X$ open as (local) sections, with global sections being elements of $\mathcal{F}(X)$.

We will also use the notation $\Gamma(U, \mathcal{F})$ and $H^0(U, \mathcal{F})$ for $\mathcal{F}(U)$.

Example 1.3. Let X be a topological space. C_X which associates to $U \subseteq X$ open the set $C_X(U)$ of continuous functions from U to $\mathbb{R}$ and restriction maps given by restriction of continuous functions defines a presheaf on X.

Example 1.4. Let X, Y be topological spaces. $\mathcal{F}$ which associates $U \subseteq X$ to the set $\mathcal{F}(U)$ of continuous functions from U to Y and restriction maps given by restriction of continuous functions defines a presheaf on X.

Example 1.5. Let X be a topological space and G an Abelian group. We can consider the association taking each open set $U \subseteq X$ to G and restriction maps the identities. This construction defines a presheaf on X.

We can alternatively define presheaves as a type of functor from the category of open sets of a topological space to C. Recall here the definition of the category of open sets of a topological space Opens_X.

Definition 1.6 (Category of Open Sets)**.** Let X be a topological space. The category Opens_X of open sets of X has objects open sets of X and morphisms inclusions.

Remark 1.7. Spelling things out, for U, V open in X there is one morphism from V to U if $V \subseteq U$ and no morphism otherwise.

We can show that C-valued presheaves are merely contravariant functors from Opens_X to C, which is essentially a formal result.

Proposition 1.8. Let X be a topological space. The data of a presheaf on X corresponds uniquely to a functor $\mathsf{Opens}_X^{\mathsf{Opp}} \to \mathsf{C}$.

Proof. This statement follows by unwinding the definitions of functoriality. A functor associates $\mathsf{Opens}_X^{\mathsf{Opp}} \to \mathsf{C}$ associates to each open set of X an object of C and by construction of the morphisms in Opens_X there is a morphism $V \to U$ if and only if $V \subseteq U$ corresponding to a morphism $U \to V$ in $\mathsf{Opens}_X^{\mathsf{Opp}}$ inducing the corresponding restriction maps in C by functoriality. Furthermore, commutativity of the triangle is preserved since the action of functors on morphisms distributes over composition and composition is unqiue. ■

One example of special interest in the study of algebraic geometry is the construction of a canonical presheaf on the spectrum of a commutative ring A.

Definition 1.9 (Spectrum of a Ring)**.** Let A be a ring. The spectrum $\mathrm{Spec}(A)$ of A is the topological space with points given by prime ideals of A and closed sets of the form $V(a)$ for $a \in A$ consisting of the prime ideals containing a.

Note that no prime ideals contain the multiplicative unit of A so for $a_1, \dots, a_n$ generating A as an A-module, the sets $D(a_i) = \mathrm{Spec}(A) \setminus V(a_i)$ consisting of the prime ideals not containing A form a basis for the toplogy of $\mathrm{Spec}(A)$.

We can now consider the following construction.

Proposition 1.10. Let A be a ring. Consider the association

$$U \mapsto \left\{ s : U \to \coprod_{\mathfrak{p} \in U} A_{\mathfrak{p}} : \begin{smallmatrix} \forall \mathfrak{p} \in U, s(\mathfrak{p}) \in A_{\mathfrak{p}} \text{ and} \\ \exists V \subseteq X, p \in V \subseteq U, \exists a, b \in A \text{ s.t. } b \notin \mathfrak{q} \forall \mathfrak{q} \in V, s(\mathfrak{q}) = \frac{a}{b} \in A_{\mathfrak{q}} \end{smallmatrix} \right\} \tag{1.2}$$

and for $V \subseteq U \subseteq X$ the forgetful maps. This association defines a presheaf of rings on $\mathrm{Spec}(A)$.

This defines a presheaf in terms of compatible stalks following [Har83, §2.2], which to me seems more artificial than in terms of localizations as in [Vak24, §4.1]

Proof. The category of rings is bicomplete, in particular admitting colimits and thus coproducts giving the sets of (1.2) the structure of a ring by pointwise operations. We get forgetful maps

$$\left\{ s : U \to \coprod_{\mathfrak{p} \in U} A_{\mathfrak{p}} : \begin{smallmatrix} \forall \mathfrak{p} \in U, s(\mathfrak{p}) \in A_{\mathfrak{p}} \text{ and} \\ \exists V \subseteq X, p \in V \subseteq U, \exists a, b \in A \text{ s.t. } b \notin \mathfrak{q} \forall \mathfrak{q} \in V, s(\mathfrak{q}) = \frac{a}{b} \in A_{\mathfrak{q}} \end{smallmatrix} \right\}$$
$$\to \left\{ s|_{U'} : U' \to \coprod_{\mathfrak{p} \in U'} A_{\mathfrak{p}} : \begin{smallmatrix} \forall \mathfrak{p} \in U', s(\mathfrak{p}) \in A_{\mathfrak{p}} \text{ and} \\ \exists V \subseteq X, p \in V \subseteq U', \exists a, b \in A \text{ s.t. } b \notin \mathfrak{q} \forall \mathfrak{q} \in V, s(\mathfrak{q}) = \frac{a}{b} \in A_{\mathfrak{q}} \end{smallmatrix} \right\}$$

for $U' \subseteq U$ that satisfy the commutativity of the diagram (1.1) and hence defines a presheaf. ■

We will set this as the structure presheaf $\mathcal{O}_{\mathrm{Spec}(A)}$ of $\mathrm{Spec}(A)$ since it captures important geometric phenomena that we seek to capture as motivated by the following example.

Example 1.11. Let k be an algebraically closed field and A a finite type k-algebra. Consider an open $U \subseteq \mathrm{Spec}(A)$ and $s : U \to \coprod_{\mathfrak{p} \in U} A_{\mathfrak{p}}$ as in (1.2). We can construct $\overline{s} : U \cap \mathrm{mSpec}(A) \to k$ whose value on each maximal ideal $\mathfrak{m}$ is the image of $s(\mathfrak{m}) \in A_{\mathfrak{m}}$ in the quotient $A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$ which is isomorphic to k since k is algebraically closed. s is regular on U and thus the association of (1.2) can be thought of as a presheaf of regular functions.

Having discussed presheaves, we can define morphisms of presheaves as follows.

Definition 1.12 (Morphism of Presheaves)**.** Let X be a topological space and $\mathcal{F}, \mathcal{F}'$ C-valued presheaves on X. A morphism of presheaves $\phi : \mathcal{F} \to \mathcal{F}'$ consists of the data of morphisms $\phi_U : \mathcal{F}(U) \to \mathcal{F}'(U)$ in C such that the diagram

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, "\phi_U"] \arrow[d, "\mathrm{res}_{U,V}"'] & \mathcal{F}'(U) \arrow[d, "\mathrm{res}'_{U,V}"] \\ \mathcal{F}(V) \arrow[r, "\phi_V"'] & \mathcal{F}'(V) \end{tikzcd}$$

is commutative for $V \subseteq U \subseteq X$ open.

Remark 1.13. A morphism of presheaves is equivalent to the data of a natural transformation between the corresponding functors $\mathsf{Opens}_X^{\mathsf{Opp}} \to \mathsf{C}$.

This data allows us to define the category of C-valued presheaves $\mathsf{PSh}(X, \mathsf{C})$ for a topological space X, where we omit the "value-category" C when it is clear from context.

Example 1.14. In analogy to Example 1.3, let X be a topological space, C_X the presheaf of continuous functions on X, and C_X^{diff} the presheaf of differentiable functions on X. There is a natural morphism of presheaves $C_X^{\mathrm{diff}} \to C_X$ induced by the inclusions of differentiable into continuous functions on each open set.

Example 1.15. Let X be a topological space and S^1 the circle. We can define a presheaf of continuous functions valued in S^1 by exponentiating a locally continuous function $e^{2\pi i f}$ for f locally continuous. This defines a morphism of presheaves C_X to S^1-valued presheaves in analogue to Example 1.4 with $Y = S^1$.

Having discussed presheaves in some depth, we can now specialize to the case of sheaves, which are presheaves satisfying additional structure.

Definition 1.16 (Sheaves)**.** Let X be a topological space and C be a category admitting arbitrary products. A presheaf $\mathcal{F}$ on X is a sheaf if for all $U \subseteq X$ and all open covers $\{U_i\}_{i \in I}$ the sequence

See also the discussion in [Stacks, Tag 00VL].

$$\mathcal{F}(U) \longrightarrow \prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i,j \in I} \mathcal{F}(U_i \cap U_j)$$

is an equalizer where the parallel maps are given by

$$(s_i)_{i\in I} \mapsto (\mathrm{res}_{U_i,U_i\cap U_j}(s_i))_{i,j\in I} \text{ and } (s_i)_{i\in I} \mapsto (\mathrm{res}_{U_j,U_i\cap U_j}(s_j))_{i,j\in I},$$

respectively.

Remark 1.17. If C is an Abelian category, the equalizer condition can be replaced by $\mathcal{F}(U)$ being the kernel of $\prod_{i\in I}\mathcal{F}(U_i) \to \prod_{i,j\in I}\mathcal{F}(U_i\cap U_j)$ by $(s_i)_{i\in I} \mapsto (s_i|_{U_i\cap U_j} - s_j|_{U_i\cap U_j})_{i,j\in I}$.

Explicitly, the local behavior states that if there is an open set $U \subseteq X$ with an open cover $\{U_i\}_{i\in I}$, a collection of sections $(s_i)_{i\in I}$ glues to a section on U if and only if $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$ for all $i, j \in I$. This proves to be the quality of sheaves that makes them more geometric than presheaves: while functions restrict naturally, they also glue naturally which is behavior captured by sheaves but not by presheaves.

Example 1.18. Let U be a topological space and $\{U_1, U_2\}$ an open cover of U. Consider continous functions from U to $\mathbb{R}$. The sheaf axiom allows for the construction of a continuous function $f : U \to \mathbb{R}$ from continuous functions $f_1 : U_1 \to R$ and $f_2 : U_2 \to \mathbb{R}$ that agree on the intersection $U_1 \cap U_2$.

In particular, we can see that Example 1.3 and Proposition 1.10 in fact define sheaves on X and $\mathrm{Spec}(A)$, respectively.

2. Lecture 2 – 11th October 2024

See [nLab-I] and the discussion therein.

Previously, morphisms of presheaves and sheaves were defined in Definitions 1.12 and 1.16. In this way, being a sheaf is a structure on presheaves. This data determines the category of sheaves as we now define.

Definition 2.1 (Category of Sheaves)**.** Let X be a topological space. The category $\mathsf{Sh}(X)$ of sheaves on X has objects sheaves and morphisms presheaf morphisms between them.

Remark 2.2. Since morphisms of sheaves are morphisms of the underlying presheaves, the forgetful functor $\mathsf{Sh}(X) \to \mathsf{PSh}(X)$ exhibits the category of sheaves as a full subcategory of the category of presheaves.

A natural question that arises from the consideration of a full subcategory is the existence of a left adjoint, which in this context amounts to the construction of a sheaf given a presheaf that is canonical in an appropriate sense. We will show such a construction is possible by considering stalks, the definition of which we now consider.

Definition 2.3 (Stalk)**.** Let X be a topological space, $x \in X$, and $\mathcal{F}$ a presheaf on X. The stalk $\mathcal{F}_x$ of $\mathcal{F}$ at x is the colimit

$$\mathcal{F}_x = \mathrm{colim}_{x \in \subseteq U} \mathcal{F}(U).$$

Remark 2.4. Such a colimit is naturally filtered since in the category $\mathsf{Opens}_X^{\mathsf{Opp}}$ since it admits a final object – the empty set $\emptyset$. The subcategory of $\mathsf{Opens}_X^{\mathsf{Opp}}$ too is filtered by taking intersections. In these settings of sheaves over topological spaces, taking stalks preserves exactness [Stacks, Tag 04B0].

By the universal property of colimits, there is a unique map from $\mathcal{F}(U)$ to $\mathcal{F}_x$ for $x \in X$ which on application of the universal property of the product induces a map $\mathcal{F}(U) \to \prod_{x \in U} \mathcal{F}_x$ which in general is neither injective nor surjective. This map, however, is injective when $\mathcal{F}$ is a sheaf.

Alternatively, the stalk of a presheaf $\mathcal{F}$ on a topological space X at a point x can be described more explicitly as equivalence classes of sections that agree on sufficently small neighborhoods given by pairs (s, U) where $s \in \mathcal{F}(U)$ and U containing x modulo the equivalence relation $(s, U) \sim (s', U')$ if there exists $V \subseteq U \cap U'$ such that $s|_V = s'|_V$. This explicit construction of the stalk allows us to see that the stalks of a presheaf retains knowledge of the sheaf in an infinitesmally small neighborhood of the point.

Lemma 2.5. Let X be a topological space and $U \subseteq X$ open. If $\mathcal{F}$ is a sheaf, the canonical map $\mathcal{F}(U) \to \prod_{x \in U} \mathcal{F}_x$ is injective.

Proof. Suppose $s, s' \in \mathcal{F}(U)$ such that they map to the same element in the product. In which case we can pick open sets V_x for each x on which $s|_{V_x} = s'|_{V_x}$. These V_x form an open cover of x so $s = s'$ since $\mathcal{F}$ is a sheaf. ■

We now describe the process of sheafification, first proving the desired construction is a sheaf before stating the definition.

Proposition 2.6. Let X be a topological space and $\mathcal{F}$ a presheaf on X. The presheaf

$$\mathcal{F}^\sharp(U) = \left\{ (s_x)_{x\in U} : \begin{smallmatrix} \forall x\in X \text{ there is an open neighborhood } V\subseteq U \text{ of } x \\ \text{and } \sigma\in\mathcal{F}(V) \text{ such that } \forall x'\in V, s_{x'}=(\sigma,V)\in\mathcal{F}_v \end{smallmatrix} \right\}$$

is a sheaf.

Proof. Suppose $\{U_i\}_{i\in I}$ is an open cover of U and $(s_{i,x})_{x\in U_i} \in \mathcal{F}^\sharp(U_i)$ where $(s_{i,x})_{x\in U_i}$ and $(s_{j,x})_{x\in U_j}$ are such that $(s_{i,x})_{x\in U_i\cap U_j} = (s_{j,x})_{x\in U_i\cap U_j}$. This glues in $\prod_{x\in U}\mathcal{F}_x$ which is a sheaf under the natural restriction maps so consider $s = (s_x)_{x\in U}$ that restricts to $(s_{i,x})_{x\in U_i}$ and $(s_{j,x})_{x\in U_j}$. For any $x \in U$ with $x \in U_i$ we can find some $V \subseteq U_i$ containing x and $\sigma \in \mathcal{F}(V)$ such that $s_{i,x'} = (\sigma, V)$ for all $x' \in V$ where since $s_{i,x'} = s_{x'}$ we have that $s = (s_x)_{x\in U}$ satisfies the condition as well. ■

In light of Proposition 2.6, we have the following.

Definition 2.7 (Sheafification)**.** Let X be a topological space and $\mathcal{F}$ a presheaf on X. The sheafification $\mathcal{F}^\sharp$ is the sheaf

$$\mathcal{F}^\sharp(U) = \left\{ (s_x)_{x\in U} : \begin{smallmatrix} \forall x\in X \text{ there is an open neighborhood } V\subseteq U \text{ of } x \\ \text{and } \sigma\in\mathcal{F}(V) \text{ such that } \forall x'\in V, s_{x'}=(\sigma,V)\in\mathcal{F}_v \end{smallmatrix} \right\}.$$

As the construction of Proposition 2.6 and the above definition suggests, sheafification does not change stalks. This expectation in fact holds.

Proposition 2.8. Let X be a topological space and $\mathcal{F}$ a presheaf on X with sheafification $\mathcal{F}^\sharp$. For all $x \in X$, $\mathcal{F}_x = \mathcal{F}^\sharp_x$.

Proof. The map $\mathcal{F}_x \to \mathcal{F}^\sharp_x$ is injective by Lemma 2.5. It remains to show surjectivity. Let $\overline{s} \in \mathcal{F}^\sharp_x$ be a section and consider $U \subseteq X$ containing x such that $\overline{s}$ is in the equivalence class of some (s, U) with $s \in \mathcal{F}^\sharp(U)$. As such, there is an open neighborhood V of U and $\sigma \in \mathcal{F}(V)$ with $s|_V$ is in the image of σ in $\prod_{x\in V}\mathcal{F}_x$ which defines a map of an element of $\mathcal{F}_x$ to $\overline{s}$. ■

This allows us to show that sheafification exhibits the expected universal property – that of the initial sheaf admitting a map from $\mathcal{F}$.

Proposition 2.9. Let X be a topological space, $\mathcal{F}$ a presheaf on X, and $\mathcal{G}$ a sheaf on X. For any morphism of presheaves $\mathcal{F} \to \mathcal{G}$ there exists a unique morphism making the diagram

$$\begin{tikzcd} \mathcal{F} \arrow[r] \arrow[d] & \mathcal{G} \\ \mathcal{F}^\sharp \arrow[ur, dashed, "\exists!"'] & \end{tikzcd}$$

commute.

Proof. We have a commutative diagram of the form

$$\begin{tikzcd} \mathcal{F} \arrow[r] \arrow[d] & \mathcal{F}^\sharp \arrow[r] \arrow[d] & \prod_{x\in X}\mathcal{F}_x \arrow[d] \\ \mathcal{G} \arrow[r] & \mathcal{G}^\sharp \arrow[r] & \prod_{x\in X}\mathcal{G}_x \end{tikzcd}$$

which induces the factorization as $\mathcal{G}^\sharp = \mathcal{G}$ since sheaves satisfy the compatibility condition imposed by the sheafification process. ■

Adjointness, structure presheaf of $\mathrm{Spec}(A)$ is a sheaf, and stalks and sections for $\mathrm{Spec}(A)$.

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